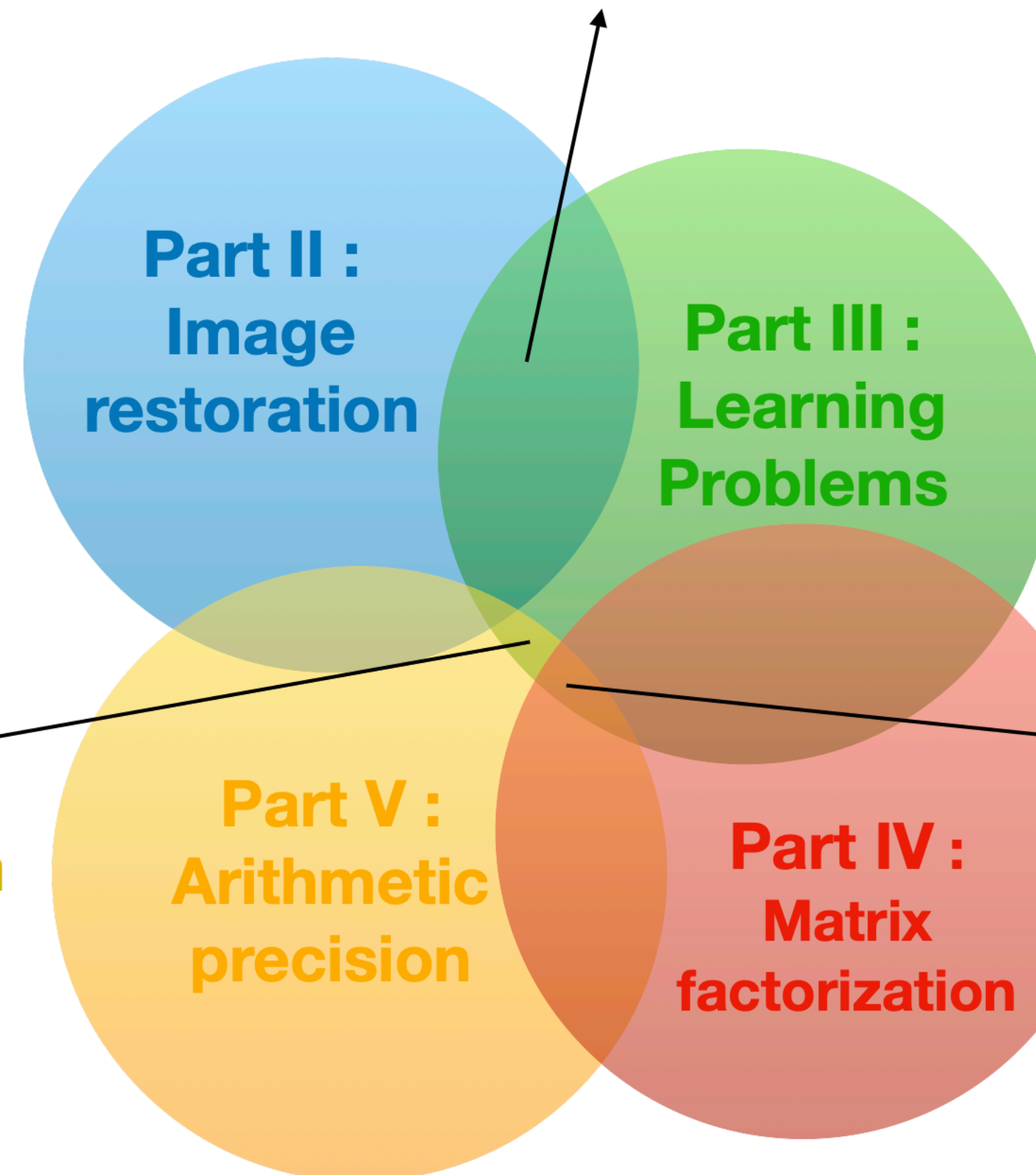


Part I : multilevel paradigm

- Higher-order order/
preconditioned non-smooth
multilevel optimization

- Convergence of the **iterates**
for multilevel methods

- Multiresolution neural-network based methods



- Rescaling invariant
quantization of
neural networks

- Mixed precision **stochastic** optimization
- Mixed precision **SciML**